

Keycap Fluid-Motion Prototyping — Bench Plan

Index-matched fluid cells · convection & magnetic drive · PDMS/wax closed cavities · Gosford bench session

Core principle: Test fluids in clear glass/plastic vessels you can see into — NOT in silicone moulds, NOT in epoxy. Run two parallel tracks (physics + casting) and marry them only after each works alone. This puts you on real physics data days sooner.

DO THIS FIRST (5 min, all on-hand): Step A1 — make a clear solid object vanish in glycerol/water. If a refractive boundary disappears on your bench today, the entire platform is real. That single test is the cheapest go/no-go for the whole dream.

PDMS — clearing the confusion

- PDMS is a **two-part cure-in-place silicone** (base + curing agent, ~10:1). You mix and cure it **solid**, exactly like your casting silicone — it is NOT poured hardener-free.
- Rule: **fully cure the PDMS first, introduce fluid second, never the reverse**. Once cured, crosslinking is complete and there is nothing to leach into the oil.
- Its appeal: refractive index ~1.40–1.43, near silicone oil — opens up clean index-matching. It also bonds/seals to itself.
- Caveat (test, don't fear): cured PDMS still *swells* in some organic solvents — tolerates silicone oil well, swells in IPA/EtOH and some oils. Covered by the soak test (B4).

On-hand inventory (confirmed)

Glycerol · baby oil · IPA · EtOH · clear epoxy (pot ~15–30 min) · casting silicone · PDMS · disc magnets (button-cell sized) · big neodymium (~50 mm) · magnetic stirrer + stir bars · mica makeup glitter (*adulterants = confound*) · polystyrene petri dishes

Need to acquire — with on-hand fallbacks so nothing blocks you

Item	Use	If you don't have it / substitute
Clear vials / cuvettes / small jars	Observation vessels	Shot glass, sample jar, cut-down clear PET bottle, perfume tester, your clear PS petri dishes (great for shallow interface + convection work)
Clear solid test object	Index-vanishing (A1)	Glass marble, clear glass cabochon/gem, borosilicate stir-rod offcut, clear glass bead, clean glass-dropper fragment, shard of PS petri dish (index ~1.59)
Silicone oil, 2 viscosities (~100 & ~350 cSt)	Oil phase + speed tuning knob	RC shock oil (sold graded by cSt — ideal), pure-dimethicone silicone lube (check label = single ingredient), or just baby oil for v1
Float-at-interface object (ρ 0.85–1.26)	A4 / A6	Nylon or acrylic craft bead; seed; peppercorn; hard-candy bead — confirm via 3-dish triage (A2)
Magnet-driven object	A6	Steel bearing/BB/steel nut , or one of your disc magnets directly
Tracer particles	Flow visualisation (A5)	Mica glitter (accept confound), printer toner (black, ~neutral buoyancy), food dye for bulk currents
Wax	Cavity former (C)	Candle, paraffin, soy wax, crayon

TRACK A Fluids & Physics — start today, no casting needed

A1 · Index-vanishing (*DO FIRST*)

Clear solid object in a vial; add glycerol, then titrate water in (or glycerol into water) until the object's edges *disappear*. Log the ratio → that's your matched aqueous index.

Success = boundary vanishes. Variant: if no glass object, use a PS petri-dish shard — higher index, may need a denser medium, but the effect is the same.

A2 · Density triage (3-dish)

Dishes of water (~1.00), baby oil (~0.85), glycerol (~1.26). Drop every candidate object in each; log float/sink:

- Floats on water → $\rho < 1.0$ → too light, scrap for interface
- Sinks in oil, floats on water → ρ 0.85–1.0 → **oil/water interface candidate**
- Sinks in water, floats in glycerol → ρ 1.0–1.26 → **glycerol-layer interface candidate**
- Sinks in glycerol → too dense (metals, glass) → **use as magnet-driven bottom-dweller instead**

Useful 0.85–1.26 band: nylon (~1.14), acrylic/PMMA (~1.18), polystyrene (~1.05), seeds, peppercorn, hard candy. Too dense: all metals, glass, ceramic, stone.

Success = ≥ 1 object that floats at a boundary.

A3 · Two-fluid interface

Layer baby/silicone oil over glycerol/water in a vial. Observe interface, then index-match across it (tune the aqueous layer) until the interface optically vanishes.

Success = visible-then-invisible interface.

A4 · Float at interface

Drop an A2 candidate into the A3 stack; tune densities (adjust glycerol ratio) until it rests at the boundary.

Success = object suspended mid-vial.

A5 · Convection

Add food dye or tracer particles. Apply a warm spot (fingertip, phone torch on a dark fleck, low desk LED). Watch for a circulating plume/cell. Compare thin vs thick fluid.

Success = you can SEE a flow cell and read streamlines.

A6 · Magnetic drive

Steel bearing or disc magnet in fluid; move the big neodymium beneath/beside. Then try the stirrer. Observe drive + **coast time** in thin vs thick oil (coast time is your slow-motion knob).

Success = controlled slow motion + measurable coast.

TRACK B Casting craft — parallel

B1 · Solid cast

Mix per ratio, pour ~12 mL in silicone mould. **Manage exotherm** (see heat box). Note pot-time, cure, demould, clarity.

B2 · Embed test

Pour part-way, wait until epoxy thickens toward end of pot life (so object neither sinks nor traps a bubble veil), place object mid-depth.

Success = object centred, clear, no bubbles. Your 15–30 min pot time gives a comfortable window for this.

B3 · Cavity former

Open cavity: greased bearing pressed/suspended in pour, removed after gel. Closed curved cavity: use Track C.

B4 · Compatibility soak (background, zero effort)

Drop a cured epoxy chip into a jar of each fluid (baby oil, silicone oil, glycerol/water, IPA). Leave 1–2 weeks. Check clouding / swelling / tackiness → tells you if a sealed cap survives long-term.

TRACK C Closed cavity via PDMS + wax (the planet/shell unlock)

1. **Shape a wax former.** Easiest first: drip molten wax into cold water (surface tension → near-spheres). Repeatable: cast wax in a tiny silicone ball mould (press a bearing into soft silicone for a hemisphere; two halves = sphere). One-off: roll softened wax between warm spoons. *Good-enough sphere beats perfect — don't over-build.*
2. **Add a fill-port.** Stick a pin / fine wire / thin wax thread from former to where the PDMS surface will be. Pull a *pin* before melt-out for a clean bore; a wax stalk melts out with the rest.
3. **Optional magic:** embed your target object (planet bead / magnet core) *inside* the wax former first. After melt-out the object is left suspended exactly where the wax held it — zero-handling placement.
4. **Mould PDMS around it; cure.** PDMS tolerates ~200°C; wax melts well below, so melt-out won't harm it.
5. **Melt / flush the wax out** through the port → closed cavity.

6. **Syringe fluids in** via the port (25–30 G needle).
7. **Seal the port:** dab uncured PDMS + re-cure, or a spot of silicone adhesive. PDMS partially self-heals fine needle tracks.
8. **Endgame:** pot the sealed PDMS cell inside an epoxy shell — PDMS chemistry + sealing, epoxy toughness.

Heat management for casting (B1) — the only place thermal matters

A 12 mL *solid* pour cures exothermically and silicone moulds insulate, so heat can't escape → risk of yellowing/bubbles/cracks. Fixes, best first:

- **Leave the back-piece off** during cure (your 80–90% covered back traps the heat). Do the mounting feature as a tiny later pour or trim it after. *Simplest fix — start here.*
- **Cool ambient room** (NOT the fridge — condensation clouds the cast and cold under-cures epoxy, leaving it tacky).
- **Gentle warmth AFTER gelling** improves clarity once the exotherm peak has passed.
- **Layer the pour** only if a single open-back pour still overheats — here it's a legitimate heat/bubble tool, not abandoning the task.

Decision gates

- **A1 fails** (can't index-match aqueous) → switch to silicone-oil matching against PDMS (~1.41); reorder strategy toward PDMS housing.
- **B4 shows clouding/swelling** → that fluid is out for sealed caps, or housing must be PDMS not epoxy.
- **A5 too sluggish / too fast** → viscosity knob: change silicone oil grade (RC shock oil cSt).
- **Marry tracks only** after A4 + A5 + A6 and B2 + B3 independently pass.

Safety — stated once

Dominant hazard = **epoxy amine sensitisation** (skin/respiratory). Covered by your existing PPE/handling protocol. **Thermal:** irrelevant for fluid cells (you're creating tiny gradients, not dissipating dangerous heat); mildly relevant only for thick *solid* casts (B1) per the heat box above. Everything else here is low-hazard.